K.Ramakrishnan College of Technology
Autonomous
Affiliated to Anna University Chennai, Approved by AICTE New Delhi, Accredited by NBA and with 'A+' grade by NAAC
Samayapuram, Tiruchirappalli - 621 112, Tamilnadu, India.
Incubating Minds, Catalyzing Careers

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Code & Name : CGI1354 & BUILDING RAG BASED SOLUTIONS

Year / Semester : III / VI

Section : A,B,C

Title of the Project : MULTI-SOURCE RAG ASSISTANCE

S. No.	Students Name	Register Number
1.	**ABDUL RAZEEK A**	**2303811710421002**
2.	**JAYASURYA L**	**2303811710421066**
3.	**KAMALESH S**	**2303811710421073**
4.	**MATHAN M**	**2303811710421092**

ABSTRACT

The Multi-Source RAG Assistance project is a Python-based desktop application that provides intelligent, context-aware answers from multiple sources like PDFs and web pages. It allows users to upload documents or enter URLs, extracting content using PyPDF, Requests, and BeautifulSoup. The extracted data is processed and structured for further use. Using LangChain, the system generates semantic embeddings, enabling efficient search and retrieval of relevant information. This forms the core of the Retrieval-Augmented Generation (RAG) pipeline. The application integrates the Gemini API to generate accurate, context-based responses while minimizing errors. A user-friendly interface is built with React framework, supporting interactive chat functionality. Additionally, Text-to-Speech (TTS) converts responses into audio for better accessibility. Overall, the project combines modern AI tools into a simple application, similar to advanced systems like NotebookLM, making it useful for education, research, and accessibility purposes.

Keywords: Retrieval-Augmented Generation (RAG) , LangChain , Semantic Embeddings , Document Processing , Gemini API , React Framework , Text-to-Speech(TTS)